Algorithm for file updates in Python

Project description

This project uses Python to update an employee IP allow list for a healthcare company. The algorithm checks the allow list against a remove list to identify IP addresses that should no longer have access. Python is used to read the file, find matching IP addresses, remove them, and save the updated list. This helps keep restricted systems secure and ensures that access is properly managed

Open the file that contains the allow list

I used the open() function with a with statement to safely open the allow_list.txt file. The with statement automatically closes the file when the code finishes using it.

```
import_file = "allow_list.txt"

with open(import_file, "r") as file:
```

Read the file contents

I used the .read() method to read all the information from the allow list and store it in the ip_addresses variable.

```
ip_addresses = file.read()
```

Convert the string into a list

I used .split() to convert the file contents from a string into a list. This makes it easier to work with individual IP addresses.

```
ip_addresses = ip_addresses.split()
```

Iterate through the remove list

I used a for loop to check each IP address in the remove_list.

```
for element in remove_list:
```

The loop allows Python to process each IP address one at a time.

Remove IP addresses that are on the remove list

I used an if statement to check whether each IP address from the remove list is in the allow list. If it is found, .remove() deletes it from the list.

```
if element in ip_addresses:
    ip_addresses.remove(element)
```

The .remove() method works correctly here because the allow list does not contain duplicate IP addresses

Update the file with the revised list of IP addresses

After removing the unwanted IP addresses, I used .join() to turn the list back into a string. The "\n" separates each IP address onto a new line. I then used .write() to save the updated list back to the file.

```
ip_addresses = "\n".join(ip_addresses)

with open(import_file, "w") as file:
    file.write(ip_addresses)
```

Summary

This algorithm automates the process of updating an IP allow list. It opens and reads the file, converts the information into a list, and checks each IP address against the remove list. Matching IP addresses are removed using an if statement and the .remove() method. Finally, the updated list is converted back into text and written to the file. This process helps maintain accurate access controls and supports better security.